

WADI AL DAWASIR, Saudi Arabia

WMO: 410610

Lat: 20.504N

Lon: 45.200E

Elev: 2062

StdP: 13.63

Time Zone: 3.00 (E03)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 46.4	(c) 49.7	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 22.8	(k) 65.2	(l) 20.8	(m) 65.5	(n) 6.5	(o) 110	(p) 0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 25.6	(c) 112.8	(d) N/A	(e) 111.3	(f) N/A	(g) 109.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 10.4	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 21.5	(b) 19.1	(c) 16.9	(d) DB	(e) 41.3	(f) 115.6	(g) 3.0	(h) 1.6	(i) 39.2	(j) 116.7	(k) 37.4	(l) 117.7	(m) 35.8	(n) 118.6	(o) 33.6	(p) 119.8	(q) N/A
(4) 21.5	(5) 19.1	(6) 16.9	(7) WB	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A	(19) N/A	(20) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.6	63.8	69.6	76.7	84.5	92.2	95.5	97.7	97.3	91.5	82.5	73.3	65.8
	DBStd	12.69	5.45	5.96	5.60	4.85	3.13	2.40	2.68	2.28	3.32	3.44	4.15	5.28
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	171	89	23	1	0	0	0	0	0	0	0	3	55
	CDD50	11899	429	549	827	1036	1307	1366	1479	1467	1244	1007	700	488
	CDD65	6595	53	152	363	586	842	916	1014	1002	794	542	253	78
	CDH74	104091	615	1650	4202	8175	13973	16084	18171	17805	12987	7145	2485	799
	CDH80	70658	148	659	2097	4923	9663	11800	13712	13348	8942	4183	991	192
(14) Wind	WSAvg	8.3	9.4	9.8	10.3	9.5	7.8	7.0	7.5	7.2	6.5	6.8	8.7	8.9
(15) Precipitation	PrecAvg	1.4	0.1	0.1	0.5	0.5	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecMax	3.5	0.8	0.7	1.9	2.5	1.9	0.0	0.3	0.2	0.0	0.3	0.6	0.6
	PrecMin	0.3	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.9	0.2	0.2	0.4	0.7	0.4	0.0	0.0	0.0	0.0	0.1	0.2	0.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.9	95.3	101.9	105.4	109.8	113.2	114.9	113.4	111.1	102.3	93.3	89.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	84.1	91.3	97.0	102.5	107.9	111.4	113.1	111.6	109.0	100.1	90.7	84.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	80.7	87.8	93.6	100.6	106.2	109.7	111.4	111.0	107.2	98.2	88.2	82.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	77.1	84.1	91.0	98.5	105.4	109.0	109.8	109.5	105.4	96.4	86.1	78.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	23.7	25.1	24.7	24.6	25.1	26.9	25.6	25.7	27.4	27.0	24.4	24.1
		MCDBR	28.5	30.0	28.8	27.4	27.2	28.2	26.6	27.0	28.6	28.0	26.7	26.6
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.484	0.528	0.597	0.625	0.640	0.664	0.682	0.670	0.574	0.467	0.467	0.451
	taud		1.842	1.730	1.595	1.578	1.533	1.481	1.492	1.510	1.655	1.872	1.882	1.909
	Ebn at Noon		246	243	233	228	223	216	212	216	235	257	248	250
	Edh at Noon		61	72	86	88	91	95	95	93	80	62	58	55
(44) All-Sky Solar Radiation	RadAvg		1554	1820	2045	2205	2356	2403	2291	2251	2161	1994	1666	1506
	RadStd		94	67	81	89	89	66	72	49	34	45	49	81

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.72	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page